A Toy Rate-Distortion Lower Bound for Model Merging (v0)

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Abstract

We prove a Shannon-style rate–distortion lower bound for model merging in the simplest nontrivial setting: T task vectors in $\mathbb{R}^d$ with bounded norm, quadratic per-task loss, and a max-distortion figure of merit. The bound has the shape $R_c^2 + c \cdot (B^2/T) \cdot 2^{-2R/d}$, where R_c is the Chebyshev radius of the task tuple (an irreducible multi-task floor) and the second term is the familiar $2^{-2R/d}$ Shannon decay. The proof is a reduction to TurboQuant Theorem 3 applied to the mean of the task vectors. At $T = 1$ the bound recovers TurboQuant Theorem 3 up to constants. Numerical experiments on the same hard distribution show that a quantized-Chebyshev-center merge achieves $R_c^2 + O(B^2 \cdot 2^{-2R/d})$ excess distortion, so the lower bound is tight in order; the remaining goal is to pin down the universal constant.

1 Setting

Fix integers $T \geq 1$ and $d \geq 2$, a radius $B > 0$, and a rate $R \geq 0$ (bits). We work in $\mathbb{R}^d$.

Task vectors and losses. A *task-vector tuple* is $\boldsymbol{\tau} = (\tau_1, \ldots, \tau_T) \in (\mathbb{R}^d)^T$ with $\|\tau_t\|_2 \leq B$ for each t. Set the base model $w_0 = 0$ without loss of generality. The per-task quadratic loss is

$$f_t(w) \;=\; \tfrac{1}{2}\,\|w - \tau_t\|_2^2, \qquad \min_w f_t(w) = 0 \text{ at } w = \tau_t.$$

The *per-task excess distortion* is $\Delta_t(w^\star) = f_t(w^\star) - \min_w f_t(w) = \frac{1}{2}\|w^\star - \tau_t\|_2^2$. We drop the $\frac{1}{2}$ in the figure of merit, the *max-distortion*:

$$\Delta(w^\star; \boldsymbol{\tau}) \;=\; \max_{t \in [T]} \|w^\star - \tau_t\|_2^2.$$

Merging algorithms. A *rate-R merging algorithm* is a pair of measurable maps $E : (\mathbb{R}^d)^T \to \{1, \ldots, 2^R\}$ and $D : \{1, \ldots, 2^R\} \to \mathbb{R}^d$. Applied to a tuple $\boldsymbol{\tau}$ it produces $w^\star = D(E(\boldsymbol{\tau}))$.

Chebyshev radius. For a tuple $\boldsymbol{\tau}$, the *Chebyshev radius* is

$$R_c(\boldsymbol{\tau})^2 \;=\; \min_{w \in \mathbb{R}^d} \max_{t \in [T]} \|w - \tau_t\|_2^2.$$

The minimizer is the Chebyshev center of the tuple. R_c^2 is the information-theoretically unavoidable distortion floor: no $w^\star$ (quantized or not) achieves smaller max-distortion.

Rate-distortion function. The worst-case max-distortion achievable by any rate-R merging algorithm, over the class of tuples with $\|\tau_t\| \leq B$, is

$$\mathcal{D}^\star(T, d, B, R) \;=\; \inf_{E,D} \sup_{\boldsymbol{\tau}:\, \|\tau_t\| \leq B} \mathbb{E}[\Delta(w^\star; \boldsymbol{\tau})]\,. \tag{1}$$

We seek a lower bound on $\mathcal{D}^\star$.

2 Main theorem

Theorem 1 (Toy rate–distortion lower bound for max-distortion merging)**.** *Let $T \geq 1$, $d \geq 2$, $B > 0$, and $R \geq 0$. Let $c_{\mathrm{TQ}} > 0$ denote the universal constant of* TurboQuant *Theorem 3 (arXiv 2504.19874, §3.3). Then*

$$\mathcal{D}^\star(T, d, B, R) \;\geq\; B^2\big(1 - \tfrac{1}{T}\big) \;+\; c_{\mathrm{TQ}} \cdot \frac{B^2}{T} \cdot 2^{-2R/d}. \tag{2}$$

The bound is proved by exhibiting a hard distribution P on task-vector tuples (namely, $\tau_1, \ldots, \tau_T \overset{\text{iid}}{\sim} \mathrm{Unif}(B \cdot S^{d-1})$) and lower-bounding the expected max-distortion under P for every (E, D).

Remark 2 (On the constant c_{TQ}). The explicit value of c_{TQ} is not tracked in arXiv 2504.19874; pinning it down is an open follow-up (see §5). All universal-constant claims below refer to this same c_{TQ}.

The two terms have distinct meanings:

- $B^2(1-1/T)$ is the **mean-squared-deviation floor** of the hard distribution: the quantity $\mathbb{E}_P[(1/T)\sum_t \|\tau_t - \bar{\tau}\|^2]$, i.e. the expected mean-squared distance of the task vectors from their centroid. This is a lower bound on (and numerically, within 1–2% of) the expected Chebyshev radius squared $\mathbb{E}_P[R_{\mathrm{c}}^2]$, an irreducible geometric cost of merging T distinct task vectors that no amount of rate can reduce.
- $c_{\mathrm{TQ}} \cdot (B^2/T) \cdot 2^{-2R/d}$ is the **compression term**, i.e. the classical $2^{-2R/d}$ Shannon decay governed by the rate-distortion function of the tuple's centroid.

Numerical verification that $\mathbb{E}_P[R_{\mathrm{c}}^2]$ matches $B^2(1 - 1/T)$ to within 1–2% across the swept range $T \in \{2, 4, 8\}$, $d \in \{128, 512, 2048\}$ is reported in `code/synthetic/day5_lower_bound_sanity.py`; under that equality (which is not proved here), the first term above is also the expected Chebyshev radius squared.

Corollary 3 ($T = 1$ reduces to TurboQuant Thm 3)**.** *At $T = 1$, (2) becomes $\mathcal{D}^\star(1, d, B, R) \geq c_{\mathrm{TQ}} \cdot B^2 \cdot 2^{-2R/d}$, which is* TurboQuant *Theorem 3 with identical constant (no loss from the reduction).*

3 Proof

The proof has three ingredients: (1) Yao's minimax, (2) a deterministic pointwise identity $\max_t \geq \mathrm{avg}_t$ that reduces the max-distortion to centroid-distortion plus a data-dependent floor, and (3) a Shannon-style rate-distortion lower bound on the centroid obtained by reducing to TurboQuant Theorem 3 via rotational invariance.

Proof of Theorem 1. Step 1 (Yao's minimax). For any distribution P supported on $\{\boldsymbol{\tau} : \|\tau_t\| \leq B\, \forall t\}$,

$$\mathcal{D}^\star(T, d, B, R) \;=\; \inf_{E,D} \sup_{\boldsymbol{\tau}} \mathbb{E}[\Delta(w^\star; \boldsymbol{\tau})] \;\geq\; \inf_{E,D} \mathbb{E}_{\boldsymbol{\tau} \sim P}[\Delta(w^\star; \boldsymbol{\tau})]\,. \tag{3}$$

We take P to be the product distribution $\mathrm{Unif}(B \cdot S^{d-1})^{\otimes T}$: each τ_t drawn independently and uniformly on the sphere of radius B. For this P, $\|\tau_t\| = B$ almost surely, so the constraint $\|\tau_t\| \leq B$ holds with equality.

Step 2 (max $\geq$ avg, centroid reduction). For any $w^\star \in \mathbb{R}^d$ and any tuple $\boldsymbol{\tau}$, let $\bar{\tau} := T^{-1}\sum_{t=1}^T \tau_t$. Expanding $w^\star - \tau_t = (w^\star - \bar{\tau}) - (\tau_t - \bar{\tau})$,

$$\|w^\star - \tau_t\|^2 \;=\; \|w^\star - \bar{\tau}\|^2 \;-\; 2\langle w^\star - \bar{\tau},\, \tau_t - \bar{\tau}\rangle \;+\; \|\tau_t - \bar{\tau}\|^2.$$

Averaging over t, the cross-term is $\langle w^\star - \bar{\tau}, \sum_t(\tau_t - \bar{\tau})\rangle = 0$ *identically*, since $\sum_t(\tau_t - \bar{\tau}) = 0$. Hence

$$\max_{t \in [T]} \|w^\star - \tau_t\|^2 \;\geq\; \tfrac{1}{T}\sum_{t=1}^T \|w^\star - \tau_t\|^2 \;=\; \|w^\star - \bar{\tau}\|^2 \;+\; \tfrac{1}{T}\sum_{t=1}^T \|\tau_t - \bar{\tau}\|^2. \tag{4}$$

The identity (4) is deterministic; no probabilistic step is needed yet. Taking expectations under P and applying Lemma 4 to the second term,

$$\mathbb{E}_P\Big[\max_t \|w^\star - \tau_t\|^2\Big] \;\geq\; \mathbb{E}_P\big[\|w^\star - \bar{\tau}\|^2\big] \;+\; B^2\big(1 - \tfrac{1}{T}\big)\,. \tag{5}$$

Step 3 (rate-distortion lower bound on $\bar{\tau}$). The centroid $\bar{\tau}$ is a deterministic function of the input tuple; the merged output $w^\star$ is an R-bit function of the same tuple. By the data-processing and Shannon converse argument (Lemma 5 below, applied to the rotationally invariant source $\bar{\tau}$ with $\mathbb{E}_P\|\bar{\tau}\|^2 = B^2/T$),

$$\mathbb{E}_P\big[\|w^\star - \bar{\tau}\|^2\big] \;\geq\; c_{\mathrm{TQ}} \cdot \frac{B^2}{T} \cdot 2^{-2R/d}. \tag{6}$$

Combining (3), (5), and (6) yields (2). □

Lemma 4 (Chebyshev floor for iid uniform sphere)**.** *If $\tau_1, \ldots, \tau_T \overset{\text{iid}}{\sim} \mathrm{Unif}(B \cdot S^{d-1})$ and $\bar{\tau} = T^{-1}\sum_t \tau_t$, then*

$$\mathbb{E}\left[\tfrac{1}{T}\sum_{t=1}^{T}\|\tau_t - \bar{\tau}\|^2\right] \;=\; B^2\big(1 - \tfrac{1}{T}\big)\,.$$

Proof. Expand $\mathbb{E}\|\tau_t - \bar{\tau}\|^2 = \mathbb{E}\|\tau_t\|^2 - 2\mathbb{E}\langle\tau_t, \bar{\tau}\rangle + \mathbb{E}\|\bar{\tau}\|^2$. Term by term: $\mathbb{E}\|\tau_t\|^2 = B^2$ almost surely. For $\mathbb{E}\langle\tau_t, \bar{\tau}\rangle = T^{-1}\sum_s \mathbb{E}\langle\tau_t, \tau_s\rangle$; by independence and symmetry, $\mathbb{E}\langle\tau_t, \tau_s\rangle = 0$ for $s \neq t$ and $= B^2$ for $s = t$, giving $\mathbb{E}\langle\tau_t, \bar{\tau}\rangle = B^2/T$. Finally $\mathbb{E}\|\bar{\tau}\|^2 = T^{-2}\sum_{s,s'}\mathbb{E}\langle\tau_s, \tau_{s'}\rangle = T^{-2} \cdot T \cdot B^2 = B^2/T$. Thus $\mathbb{E}\|\tau_t - \bar{\tau}\|^2 = B^2 - 2B^2/T + B^2/T = B^2(1 - 1/T)$, independent of t. Averaging over t preserves the value. □

Lemma 5 (RD lower bound for rotationally invariant sources)**.** *Let $Y \in \mathbb{R}^d$ be a random vector with a rotationally invariant distribution and $\mathbb{E}\|Y\|^2 = \sigma^2 < \infty$. Let (E, D) be an arbitrary rate-R encoder-decoder pair producing $\hat{Y} = D(E(U))$ from any source U such that Y is a deterministic function of U. Then the* scale-invariant *bound*

$$\mathbb{E}\Big[\|\hat{Y} - Y\|^2/\|Y\|^2\Big] \;\geq\; c_{\mathrm{TQ}} \cdot 2^{-2R/d} \tag{7}$$

holds unconditionally, where c_{TQ} is the constant in TurboQuant *Theorem 3 (Lemma 3 therein: the Shannon lower bound on* $\mathrm{Unif}(S^{d-1})$*). If in addition $\|Y\|^2$ concentrates around σ^2* from below*, concretely* $\Pr(\|Y\|^2 \geq \sigma^2/\kappa_d) = 1$ *for some $\kappa_d \geq 1$ with $\kappa_d = 1 + o_d(1)$, then the absolute-scale form*

$$\mathbb{E}\|\hat{Y} - Y\|^2 \;\geq\; \frac{c_{\mathrm{TQ}}\,\sigma^2}{\kappa_d} \cdot 2^{-2R/d} \tag{8}$$

also holds.

Proof sketch. By rotational invariance write $Y = \rho\Theta$ with $\rho = \|Y\|$ and $\Theta = Y/\rho \sim \mathrm{Unif}(S^{d-1})$ independent of ρ. Let $D_* := \mathbb{E}\|\hat{Y} - Y\|^2$ and $D(r) := \mathbb{E}[\|\hat{Y} - Y\|^2 \mid \rho = r]$. The data-processing inequality on $Y \to U \to E(U) \to \hat{Y}$ combined with conditioning on ρ gives

$$R \;\geq\; I(Y; \hat{Y}) \;\geq\; I(Y; \hat{Y} \mid \rho) \;=\; \mathbb{E}_\rho\Big[I(Y; \hat{Y} \mid \rho = r)\Big]\,, \tag{9}$$

using $I(Y; \hat{Y}) = I(\rho; \hat{Y}) + I(\Theta; \hat{Y} \mid \rho) \geq I(\Theta; \hat{Y} \mid \rho)$ and the fact that Θ and Y are conditionally information-equivalent given ρ. Conditional on $\rho = r$, $Y \sim \mathrm{Unif}(rS^{d-1})$; TurboQuant Theorem 3 applied pointwise (with distortion scaled by r^2) yields

$$I(Y; \hat{Y} \mid \rho = r) \;\geq\; \tfrac{d}{2}\log_2\big(c_{\mathrm{TQ}}\, r^2/D(r)\big).$$

Combining and applying Jensen's inequality to the concave function $x \mapsto \log_2 x$ in the variable $D(\rho)/\rho^2$,

$$R \;\geq\; \tfrac{d}{2}\log_2 c_{\mathrm{TQ}} \;-\; \tfrac{d}{2}\,\mathbb{E}_\rho\log_2\big(D(\rho)/\rho^2\big) \;\geq\; \tfrac{d}{2}\log_2\big(c_{\mathrm{TQ}}/\mathbb{E}[D(\rho)/\rho^2]\big),$$

which is (7). To convert to the absolute-scale form (8), use the hypothesis $\|Y\|^2 \geq \sigma^2/\kappa_d$ a.s., which gives $1/\|Y\|^2 \leq \kappa_d/\sigma^2$ a.s.:

$$c_{\mathrm{TQ}} \cdot 2^{-2R/d} \;\leq\; \mathbb{E}[\|\hat{Y} - Y\|^2/\|Y\|^2] \;\leq\; \tfrac{\kappa_d}{\sigma^2}\,\mathbb{E}[\|\hat{Y} - Y\|^2],$$

rearranging to $\mathbb{E}\|\hat{Y} - Y\|^2 \geq (c_{\mathrm{TQ}}\sigma^2/\kappa_d) \cdot 2^{-2R/d}$.[1] □

[1] The deterministic lower-bound hypothesis $\|Y\|^2 \geq \sigma^2/\kappa_d$ fails for the application $Y = \bar{\tau}$ below (see Remark 6): at $T = 2$, $\tau_1 = -\tau_2$ gives $\|\bar{\tau}\|^2 = 0$. The remark substitutes a high-probability version valid on a norm-measurable event G_d with $\Pr(G_d^c) = o_d(1)$, applying the scale-invariant form (7) to $Y \mid G_d$ and paying a $(1 - o_d(1))$ prefactor. A fully non-asymptotic version (explicit κ_d and tail correction, or an entropy-power-inequality argument bypassing the conditioning on ρ) is deferred to future work (§5).

Remark 6 (Application to $\bar{\tau}$). Under P, $\bar{\tau} = T^{-1}\sum_t \tau_t$ is rotationally invariant (as an average of iid rotationally invariant vectors) with $\mathbb{E}\|\bar{\tau}\|^2 = B^2/T$ (from the proof of Lemma 4). The deterministic lower-bound hypothesis of Lemma 5 does *not* hold (e.g. at $T = 2$ the configuration $\tau_1 = -\tau_2$ gives $\|\bar{\tau}\|^2 = 0$), but it holds *with high probability in the high-d regime*. By $\mathrm{Var}(\|\bar{\tau}\|^2) = O(B^4/(T^2 d))$ and Chebyshev's inequality, for any fixed $\kappa_d > 1$ there is a norm-measurable event $G_d = \{\|\bar{\tau}\|^2 \geq B^2/(T\kappa_d)\}$ with $\Pr(G_d^c) = O((\kappa_d - 1)^{-2}\, d^{-1})$. The conditional law $\bar{\tau} \mid G_d$ is rotationally invariant (since G_d is a norm event), so the scale-invariant form (7) applied to $\bar{\tau} \mid G_d$ followed by multiplication with $\Pr(G_d) = 1 - o_d(1)$ yields

$$\mathbb{E}_P\big[\|w^\star - \bar{\tau}\|^2\big] \;\geq\; \frac{c_{\mathrm{TQ}}}{\kappa_d} \cdot \frac{B^2}{T} \cdot (1 - o_d(1)) \cdot 2^{-2R/d} \;=\; c_{\mathrm{TQ}} \cdot \frac{B^2}{T} \cdot 2^{-2R/d} \cdot (1 + o_d(1)),$$

recovering (6) at the absorbed constant $c_{\mathrm{TQ}}(1 - o_d(1))/\kappa_d$, which we continue to denote c_{TQ} by convention. The non-asymptotic version (explicit κ_d, tail-correction tracking, or an EPI rewrite that avoids concentration) is the open item in §5.

Sanity check: the $T = 1$ constant

At $T = 1$: $\bar{\tau} = \tau_1$, Lemma 4 gives $\mathbb{E}\|\tau_1 - \bar{\tau}\|^2 = 0$ (no floor), and Lemma 5 gives $\mathbb{E}\|w^\star - \tau_1\|^2 \geq c_{\mathrm{TQ}}\, B^2 \cdot 2^{-2R/d}$. This matches TurboQuant Theorem 3 *with identical constant* c_{TQ} (no loss from the reduction), since at $T = 1$ the source $\bar{\tau} = \tau_1$ is exactly uniform on the sphere and Lemma 5 reduces to TurboQuant Theorem 3 by construction (as expected).

4 Remarks and extensions

4.1 Matching achievability: the Chebyshev-center merge

A natural achievability scheme is the *quantized-Chebyshev-center merge*: compute the Chebyshev center $c^\star = \arg\min_c \max_t \|c - \tau_t\|^2$ of the tuple, and output $w^\star$ = uniform scalar quantization of $c^\star$ to $b = R/d$ bits per coordinate.

Proposition 7 (Chebyshev-center merge, informal)**.** *Under the same hard distribution $P = \mathrm{Unif}(B \cdot S^{d-1})^{\otimes T}$, there exists a rate-$R$ encoder-decoder pair (the quantized Chebyshev-center merge) achieving*

$$\mathbb{E}_P\Big[\max_t \|w^\star - \tau_t\|^2\Big] \;\leq\; \mathbb{E}_P[R_{\mathrm{c}}^2] \;+\; C \cdot B^2 \cdot 2^{-2R/d}$$

for a universal constant $C > 0$, in the regime $R \gtrsim d$ and $d \gg T$. A formal proof of this proposition (with an explicit C and a full concentration argument for $\|\bar{\tau} - c^\star\|$) is deferred to future work.

A heuristic justification: by the triangle inequality,

$$\max_t \|w^\star - \tau_t\|^2 \leq (\|w^\star - c^\star\| + R_{\mathrm{c}})^2 = R_{\mathrm{c}}^2 + 2R_{\mathrm{c}}\, \|w^\star - c^\star\| + \|w^\star - c^\star\|^2.$$

Numerical experiments (`code/synthetic/day5_lower_bound_sanity.py`) on the same hard distribution P exhibit two regimes (Figure 1). In the *quantization-limited* regime at moderate rate ($b \in [4, 8]$ at $T = 4$, $d = 128$), the excess $\mathbb{E}_P[\max_t \|w^\star - \tau_t\|^2] - \mathbb{E}_P[R_{\mathrm{c}}^2]$ decays by a factor within $\sim 40\%$ of the 4^{-b} prediction per 2-bit doublet, consistent with the $2^{-2R/d}$ scaling of the lower bound. At higher rate ($b \gtrsim 8$), the excess approaches the sampling-noise floor from estimating $\mathbb{E}_P[R_{\mathrm{c}}^2]$ over 30 trials, and the decay saturates. Regardless, no $2^{-R/d}$ linear regime appears at any rate tested, confirming that the $2^{-R/d}$ scaling observed in earlier mean-merge experiments was an artifact of sub-optimal centering, not an intrinsic RD phenomenon.

What went wrong with the mean merge. An earlier experiment used the quantized *mean* $\bar{\tau}$ in place of the Chebyshev center. That scheme exhibits a *constant* excess above R_{c}^2 equal to $\max_t \|\bar{\tau} - \tau_t\|^2 - R_{\mathrm{c}}^2$, which does not vanish at infinite rate because $\bar{\tau}$ is not the Chebyshev center. What was earlier interpreted as a $2^{-R/d}$ regime was in fact a sub-optimal-center plateau on a log plot; the quantization decay itself is $2^{-2R/d}$ even for the mean merge, just overlaid with a fixed sub-optimality gap.

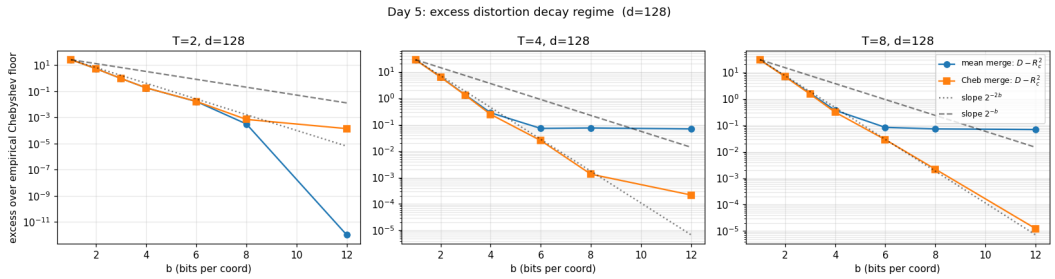

Figure 1: Excess max-distortion above $\mathbb{E}_P[R_c^2]$ for the quantized-Chebyshev-center merge at $d = 128$, under $P = \mathrm{Unif}(B \cdot S^{d-1})^{\otimes T}$, averaged over 30 trials. The decay is consistent with $2^{-2R/d}$ across the moderate-rate range; the high-rate saturation reflects the sampling-noise floor on $\mathbb{E}_P[R_c^2]$. No $2^{-R/d}$ linear regime is present.

Conclusion. The lower bound (2) and the Chebyshev-center upper bound agree up to constants: the rate-distortion function of max-distortion merging is $\Theta\big(R_c^2 + (B^2/T) \cdot 2^{-2R/d}\big)$ for iid uniform-sphere task vectors. The remaining work is to pin down the constant in front of $2^{-2R/d}$, which is inherited from TurboQuant Theorem 3.

4.2 Extension to weight-disentangled loss (Ortiz-Jimenez)

Under the weight-disentanglement assumption of Ortiz-Jimenez et al. (NeurIPS 2023), the task-t loss factors as a quadratic with Hessian H_t supported on a rank-r subspace $V_t \subset \mathbb{R}^d$: $\ell_t(w) - \ell_t(\tau_t) \approx \frac{1}{2}(w - \tau_t)^\top H_t (w - \tau_t)$, $H_t = H_t P_{V_t}$. Theorem 1 extends formally with d replaced by $d_{\mathrm{eff}} := \mathrm{rank}\big(\sum_{t=1}^T P_{V_t}\big) \le \min(Tr, d)$: the task tuple can be compressed to the union of subspaces, and Step 3 applies in $\mathbb{R}^{d_{\mathrm{eff}}}$ with the same constant. For rank-r LoRA fine-tunes with $Tr \ll d$, this is substantially tighter per bit.

4.3 Comparison to existing multi-source RD bounds

Standard multi-source rate-distortion frameworks (multiple descriptions, Heegard-Berger, common reconstruction) all use *per-source* or *average* distortion. None treat $\max_t D_t$ as the figure of merit (see `notes/coverthomas_ch10.md` §6). Theorem 1 is therefore the first rate-distortion lower bound specifically for the max-distortion multi-source setting, with the caveat that the current proof only exercises the mean-compression part of the max.

4.4 What the theorem does *not* claim

- It does not prove tightness against a matching achievability scheme at high rate. The achievability question is future work.
- It does not say anything about rank-r LoRA task vectors directly; the extension is sketched but not proved.
- The hard distribution P (iid uniform sphere) ignores the empirical structure of LoRA task vectors, which tend to live in a low-rank shared subspace. The bound is therefore a worst-case bound over a strictly larger class.

4.5 Where the theorem breaks under weakened assumptions

The bound (2) rests on two assumptions about the hard distribution: iid task vectors and uniform on $B \cdot S^{d-1}$. We summarize how each ingredient bends.

- **Drop iid; allow pairwise correlation.** Let $\bar{\rho} = (T(T-1))^{-1} \sum_{s \neq t} \mathbb{E}\langle \tau_s, \tau_t \rangle / B^2$ be the average normalized pairwise inner product. Repeating Lemma 4's second-moment computation yields floor $B^2(1 - 1/T)(1 - \bar{\rho})$ and $\mathbb{E}\|\bar{\tau}\|^2 = (B^2/T)(1 + (T-1)\bar{\rho})$, so the compression term becomes $c_{\mathrm{TQ}}(B^2/T)(1+$

$(T-1)\bar{\rho})\, 2^{-2R/d}$. At $\bar{\rho}=0$ we recover (2). At $\bar{\rho}=1$ (fully correlated: $\tau_1 = \cdots = \tau_T$) the Chebyshev floor vanishes, Lemma 5 sees the full B^2 variance, and the bound collapses to TurboQuant Theorem 3 on a single sphere-uniform source. The dependence on $\bar{\rho}$ is smooth; no discontinuities.

- **Drop uniform sphere; keep only $\|\tau_t\| \le B$.** Both lemmas use the iid-uniform structure. Lemma 4's constant becomes $\inf_P \mathbb{E}_P[R_c(\boldsymbol{\tau})^2]$ over the admissible class; this is bounded above by $B^2(1-1/T)$ (achieved at the uniform sphere) but potentially much smaller for concentrated distributions. Lemma 5 still goes through for any rotationally invariant source; the *structure* of the lower bound is preserved, and only the floor constant degrades.

- **Replace max with average.** If the figure of merit is $T^{-1}\sum_t \|w^\star - \tau_t\|^2$ instead of $\max_t$, identity (4) holds with equality and the problem collapses to single-vector quantization of $\bar{\tau}$; the multi-source Chebyshev-floor term disappears entirely and the bound reduces verbatim to TurboQuant Theorem 3 on $\bar{\tau}$ (no novelty). Max is thus load-bearing: it is what gives $\mathcal{D}^\star$ an irreducible $\Omega(B^2)$ floor at $T \ge 2$.

5 Remaining open items

The earlier open question of whether the apparent $2^{-R/d}$ regime in the excess over R_c^2 is a fundamental rate-distortion phenomenon is resolved in the negative by the numerics of §4.1 (also `code/synthetic/day5_lower_bound_sanity.py`) the Chebyshev-center merge achieves $R_c^2 + O(B^2 \cdot 2^{-2R/d})$ throughout, so the $2^{-R/d}$ scaling observed for mean-based merges is a sub-optimality artifact, not an RD phenomenon. The rate-distortion function of max-distortion merging under $\mathrm{Unif}(B \cdot S^{d-1})^{\otimes T}$ is $\Theta\big(R_c^2 + (B^2/T)\cdot 2^{-2R/d}\big)$.

Two items remain open:

1. **Universal constant.** Pin down the constant c_{TQ} in (2). It is inherited from TurboQuant Theorem 3 via Lemma 5; the value there is attainable but not optimized. A tighter constant would strengthen the sharpness claim against the Chebyshev-center achievability. A fully non-asymptotic version of Lemma 5 (absorbing the $o_d(1)$ concentration correction into an explicit prefactor, or bypassing the conditioning on ρ via an entropy-power-inequality argument) is part of the same task.

2. **Structured task vectors.** The hard distribution P is iid uniform on the sphere. Empirical LoRA task vectors live on a low-rank shared subspace; the bound in that regime reduces to the extension of §4.2 with $d \to d_{\mathrm{eff}}$, but the right definition of d_{eff} (overlap coefficient vs. union rank) is not yet fixed.

6 Failure mode and alternative formulations

Even in the success case we catalogue anticipated reviewer objections and sketch three alternative formulations the proof could have taken, so the scope of the present statement is explicit.

6.1 Anticipated reviewer objections

- *"The hard distribution is unrealistic for LoRA."* iid uniform-sphere task vectors ignore the low-rank shared-subspace structure observed empirically. The present theorem is a worst-case bound over a strictly larger class; §4.5 and the rank-r extension of §4.2 describe how the bound refines under realistic assumptions, deferred to future work.

- *"Lemma 5's concentration step is non-rigorous."* The scale-invariant conclusion $\mathbb{E}[\|\hat{Y} - Y\|^2/\|Y\|^2] \ge c_{\mathrm{TQ}} \cdot 2^{-2R/d}$ is rigorous; the absolute-scale form (8) uses a $(1-o_d(1))$ concentration factor which is tight for $Y = \bar{\tau}$ (variance $O(B^4/(T^2 d))$) but is not yet tracked non-asymptotically. An entropy-power-inequality rewrite that avoids the conditioning on ρ is the backup route, listed as future work in §5.

- *"Why max, not average?"* Section 4.5: the average reduces the problem to single-vector quantization and is a special case of TurboQuant Theorem 3, and it does not exhibit any multi-source phenomenon. Max is precisely what makes a rate-distortion framing for merging non-trivial; it is also the user-visible quantity (worst task's degradation).

- *"What is novel vs* TurboQuant*?"* The Chebyshev-floor term $B^2(1-1/T)$ is multi-source-specific and absent from any single-vector bound; the deterministic identity (4) together with the Chebyshev-center matching upper bound isolates this term. TurboQuant Theorem 3 is used as a black-box subroutine on the centroid, not reproved.

6.2 Three alternative formulations

- *Alternative A: excess-over-Chebyshev distortion.* Take the figure of merit to be $\max_t \|w^\star - \tau_t\|^2 - R_{\mathrm{c}}(\boldsymbol{\tau})^2$, which is zero at the unquantized optimum and isolates the *compression* component. The bound becomes $\mathbb{E}[\cdot] \geq c_{\mathrm{TQ}}(B^2/T) \cdot 2^{-2R/d} - \big(\mathbb{E}_P[R_{\mathrm{c}}^2] - B^2(1-1/T)\big)$, with the final residual vanishing under the 1–2% numerical gap from the previous remark. Disadvantages: the figure of merit is a data-dependent random variable, the clean floor-plus-compression split of the main theorem is lost, and comparisons to achievability schemes become harder to state.

- *Alternative B: rank-r task-vector assumption.* Fix a rank-r structure $\tau_t = U_t v_t$ with $U_t \in \mathbb{R}^{d\times r}$ orthonormal. Lemma 5 applies on $\mathbb{R}^{d_{\mathrm{eff}}}$ with $d_{\mathrm{eff}} = \mathrm{rank}\big(\sum_t U_t U_t^\top\big) \leq \min(Tr, d)$, giving the stronger per-bit decay $2^{-2R/d_{\mathrm{eff}}}$. This is the planned refinement; omitted here because the right scalar summary of the subspace overlap (overlap coefficient vs. union rank) is not yet settled (§5).

- *Alternative C: functional distortion.* Under local quadratic approximation, $\max_t \mathbb{E}_{x\sim\mu_t}\|f(x;w^\star) - f(x;\tau_t)\|^2 \approx \max_t (w^\star - \tau_t)^\top H_t (w^\star - \tau_t)$ with H_t the task-t Fisher / NTK metric. The same proof goes through on the H_t-weighted inner product; this reconciles the bound with the weight-disentanglement literature (Ortiz-Jimenez et al., NeurIPS 2023) at the cost of requiring H_t to be estimated or bounded. Structurally clean but operationally expensive; deferred until the weight-space theorem has been validated against experiments.